QQ-A-250/15E
January 29, 1971
SUPERSEDING
Fed. Spec. QQ-A-250/15D
March 17, 1967

FEDERAL SPECIFICATION

ALUMINUM ALLOY ALCLAD 7178, PLATE AND SHEET

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

(This specification forms a part of the latest issue of Federal Specification QQ-A-250.)

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers the specific requirements for aluminum alloy alclad 7178 plate and sheet; the general requirements are covered in QQ-A-250. The plate and sheet covered by this specification shall be an integral composite product consisting of a heat-treatable aluminum alloy (7178) core with thin layers of an aluminum alloy (7072) anodic to the core and of approximately equal thickness bonded to both surfaces.

1.2 Classification.

1.2.1 Tempers. The plate and sheet shall be classified as O, T6, T62, T651 or F temper, as specified (see 6.2). The definitions of these tempers shall be as specified in American National Standards Institute ANSI Std. H35.1.

2. APPLICABLE DOCUMENTS

Latest issue of Federal Specification:

QQ-A-250 - Aluminum and Aluminum Alloy Plate and Sheet; General Specification for.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC, 20402.

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(Single copies of this specification and other Federal specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, D. C., Atlanta, Chicago, Kansas City, Mo., Fort Worth, Denver, San Francisco, Los Angeles and Seattle, Washington.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from the established distribution points in their agencies.)

Military Specification:

MIL-I-8950 - Inspection, Ultrasonic, Wrought Metals, Process for.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

American National Standards Institute ANSI Standard:

H35.1 - Alloy and Temper Designation Systems for Aluminum.

(Application for copies should be addressed to the American National Standards Institute, 1430 Broadway, New York, NY 10018.)

3. REQUIREMENTS

3.1 Chemical composition.

3.1.1 The chemical composition of the core ingots or slabs and of the cladding plates used for the manufacture of the alclad plates and sheets shall conform to the requirements shown in table I for core and cladding, respectively.

TABLE I. Chemical composition 1/

Element	Analysis			
	Core (7178)		(Cladding (7072)	
	Minimum	Maximum	Minimum	Maximum
	Percent	Percent	Percent	Percent
Zinc	6.3	7.3	0.8	1.3
Magnesium	2.4	3.1	-	0.10
Copper	1.6	2.4	-	.10
Chromium	0.18	0.35	-	-
Manganese	-	.30	-	.10
Iron	-	.50	-	2/
Silicon	-	.40	-	2/
Titanium	-	.20	-	-
Others, each	-	.05	-	.05
Others, total	-	.15	-	.15
Aluminum	Remainder		Remainder	

1/ Analysis shall regularly be made only for the elements specifically mentioned in the above table. If, however, the presence of other elements is indicated in the course of routine analysis, further analysis shall be made to determine conformance to the limits specified for other elements.

2/ Iron plus silicon 0.7 percent maximum.

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3.2 Mechanical properties.

3.2.1 Mechanical properties of material as supplied. The mechanical properties perpendicular to the direction of final rolling, except for material less than 9 inches in width, shall conform to the requirements of table II for the temper specified. For material less than 9 inches in width, the mechanical properties parallel to the direction of final rolling shall conform to the requirements of table II for the temper specified.

TABLE II. Mechanical properties

Temper	Thickness	Tensile strength, minimum	Yield strength at 0.2 percent offset, minimum	Elongation in 2 in. or 4 times D 1/, 2/, minimum
	Inches	P.s.i.	P.s.i.	Percent
O	0.015 thru 0.062	3/ 36,000	3/ 20,000	10
	.063 thru .187	3/ 38,000	3/ 20,000	10
	.188 thru .499	3/ 40,000	3/ 21,000	10
T6 and T62 5/	0.015 thru 0.044	76,000	66,000	7
	.045 thru .062	78,000	68,000	8
	.063 thru .187	80,000	70,000	8
	.188 thru .249	82,000	71,000	8
-T651 and T62 5/	0.250 thru 0.499	82,000	71,000	8
	4/ .500 thru 1.000	84,000	73,000	6
	4/ 1.001 thru 1.500	84,000	73,000	4
	4/ 1.501 thru 2.000	80,000	70,000	3
F	All	6/	6/	6/

1/ Not required for material 1/2 inch or less in width.
2/ D represents specimen diameter.
3/ Maximum
4/ These properties are those of the core material inasmuch as the tests are made on a round specimen machined from the plate. The minimum tensile and yield strength across the whole section of the plate shall be about 2.5 percent lower than these values.
5/ Material in the T62 temper is not available from the materials producers.
6/ No requirements.

3.2.2 Mechanical properties after heat treatment. In addition to conforming to the requirements of 3.2.1, material in the annealed (O) and the as-fabricated (F) tempers shall, after proper solution heat treatment and aging, also conform to the requirements of table II for the T6 and T62 tempers. Material as received in the T6 and T651 tempers shall, after proper re-solution heat-treatment and aging, be capable of conforming to the requirements specified in table II for the T6 and T62 tempers.

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3.3 Cladding thickness.

3.3.1 Thickness of cladding plates. The aluminum alloy (7072) plates that are bonded to the two sides of the aluminum alloy (7178) ingot or slab to form a composite that is to be rolled to the finished thickness shall each have a thickness as specified in table III.

TABLE III. Cladding thickness

Thickness of finished plate or sheet	Nominal cladding thickness per side; percent of composite thickness	Average thickness per side of cladding on finished plate or sheet; minimum percent of plate or sheet thickness
Inches		
Under 0.063	4	3.2
0.063 thru 0.187	2.5	2
0.188 and over	1.5	1.2 1/

1/ For plate 0.500 and over, the average cladding thickness per side shall have a maximum value of 3 percent of the plate thickness.

3.3.2 Thickness of cladding. If question arises concerning the thickness of cladding of the finished sheet or plate, samples examined in accordance with QQ-A-250 shall show an average of cladding on each side, not less than that specified in table III.

3.4 Internal defects. When specified (see 6.2), plate shall be ultrasonically inspected (see QQ-A-250). Acceptance limits shall be as specified in table IV.

TABLE IV. Ultrasonic discontinuity acceptance limits 1/

Thickness	Maximum weight per piece	Discontinuity class 2/
Inches	Pounds	
0.500 thru 1.499	2,000	B
1.500 thru 2.000	2,000	A

1/ Discontinuities in excess of those listed in table IV may be allowed subject to the approval of the procuring activity, if it is established that they will be removed by machining or that they are in noncritical areas.
2/ The discontinuity class limits are defined in MIL-I-8950.

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3.5 Marking. In addition to the marking required in Fed. Std. No. 184, material in the T6 and T651 tempers shall be identified by a lot number marked in at least one location on each piece.

4. QUALITY ASSURANCE PROVISIONS
(See QQ-A-250.)

4.1 Mechanical tests after heat treatment.

4.1.1 Number of tests, after heat treatment. From material in the annealed (O) and as-fabricated (F) tempers, an additional number of specimens equal to that required by QQ-A-250 shall be taken and tested after solution heat treatment and artificial aging to determine compliance with 3.2.2.

5. PREPARATION FOR DELIVERY
(See QQ-A-250.)

6. NOTES

6.1 Intended use. The material covered by this specification is intended for use in high-strength aeronautical applications. This material has poor resistance to fracture propagation. This factor should be considered in the design of structures for weapons systems.

6.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

(a) Title, number, and date of this specification.

(b) Form and temper required (see 1.2.1).

(c) Dimensions required.

(d) Requirements for sizes not specifically covered (see QQ-A-250).

(e) Whether ultrasonic inspection of plate is required (see 3.4).

(f) Selection of applicable levels of preservation, packaging, and packing required, if other than level C (see QQ-A-250).

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6.3 Temper. Temper T651 is available in plate only.

MILITARY CUSTODIANS:
Army - MR
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Review activities:
Army - MR, MI, MU, WC
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Army - EL, ME
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SPECIFICATION ANALYSIS SHEET

Form Approved
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INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION
QQ-A-250/15E ALUMINUM ALLOY ALCLAD 7178, PLATE AND SHEET

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

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